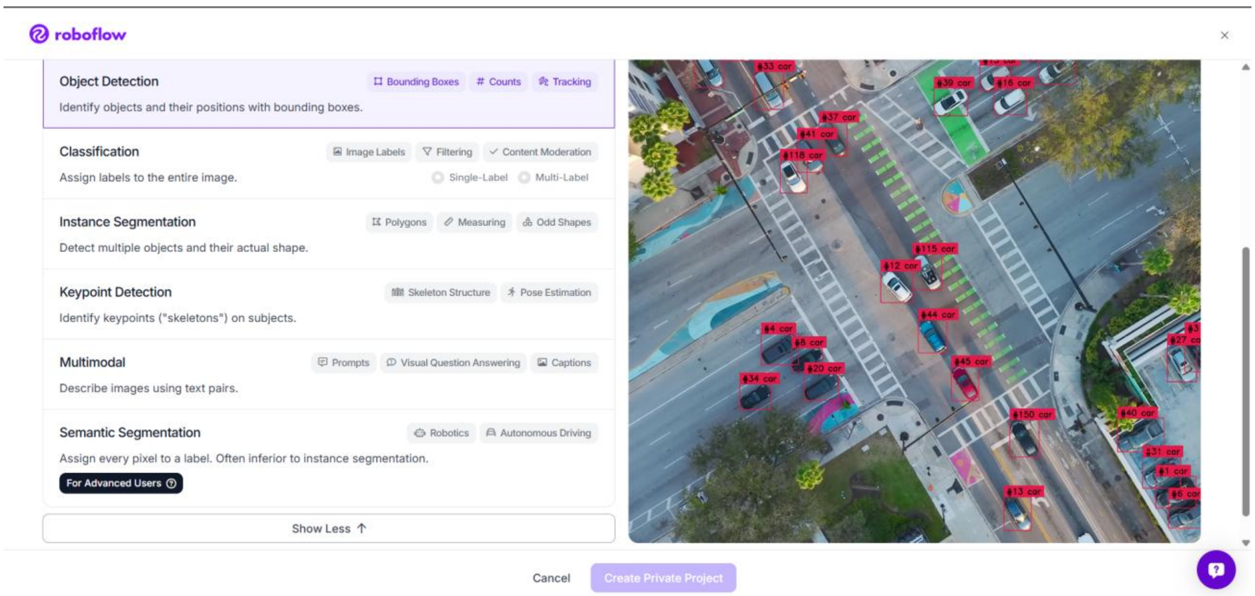
WEEK 3: DATASET ENGINEERING, LABEING SYSTEM & METHODOLOGY TRIAGE

During this phase of the Undergraduate Project 1 (UGP-1) cycle, focus was directed toward importing, testing, and scoping our primary fieldwork imagery dataset (collected from the areas affected by Ditwa 2025 near Kandy, Sri Lanka) within the Roboflow computer vision platform.

A sandbox prototype environment was initialized under the workspace designated as "sample_project_3" to evaluate how different annotation geometries impact both the downstream Machine Learning accuracy and the upstream Semantic Communication bandwidth overhead.

1. ROBOFLOW WORKSPACE SETUP & INITIAL TESTS

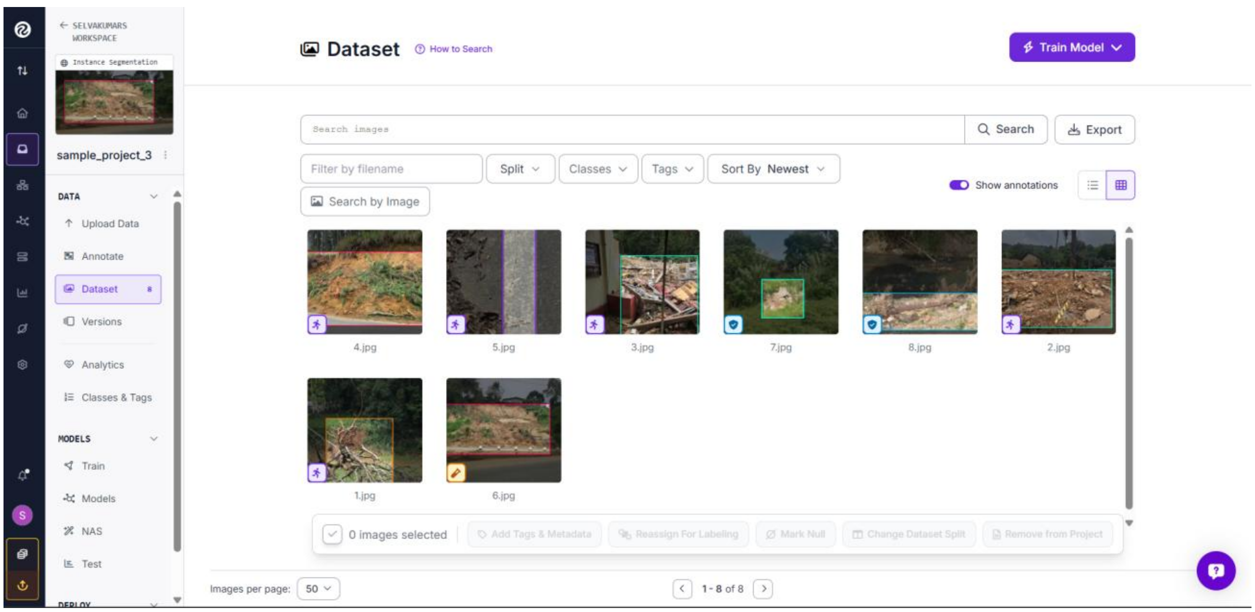
The environment was established to process custom high-resolution aerial and ground-level fieldwork imagery. Preliminary iterations involved testing multimodal annotation options to determine the most stable baseline structure.



2. ANNOTATION TRIAGE: OBJECT DETECTION VS. CLASSIFICATION

Initial annotation efforts focused heavily on Object Detection methodologies utilizing bounding boxes. Features such as structural debris, landslides, and uprooted trees were manually isolated with rectangular coordinates.

However, during this process, technical doubts arose regarding the efficiency of coordinate-heavy annotations when paired with a semantic data pipeline.



3. ADVISOR CONSULTATION & STRATEGIC DECISION LOG

A formal project consultation was held with supervisor Dr. Himal Suraweera to resolve architectural ambiguities.

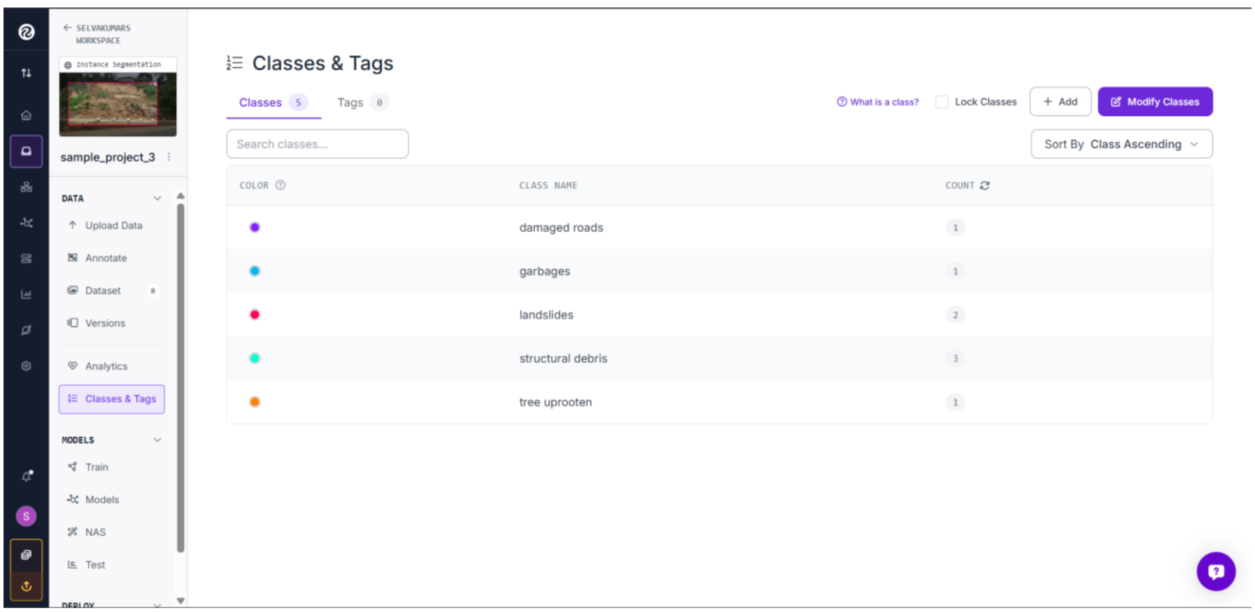
• Technical Query Raised: Which annotation methodology (Object Detection bounding boxes or Full-Frame Image Classification) yields optimal performance within an edge-deployed semantic communication system?

• Supervisor Guidance: Dr. Himal explicitly advised pivoting the core pipeline framework toward Image Classification.

• Engineering Justification:

In a crippled, bandwidth-constrained post-disaster network environment, transmitting continuous arrays of bounding box coordinate matrices ([x_center, y_center, width, height]) incurs severe data overhead and channel noise vulnerability. Conversely, an Image Classification network compresses the scene into a singular, highly compact semantic intent vector (class probability distribution array). This pivot massively reduces transmission payloads (Kilobytes instead of Megabytes), ensuring successful assessment deliveries over degraded channels.

4. CRITICAL CHALLENGE: PRIMARY DATASET CLASS IMBALANCE



A localized inventory of the uploaded Kandy fieldwork dataset highlighted an immediate class distribution challenge. While specific disaster expressions are well-represented, other critical indicators remain structurally sparse:

• Well-Represented (Strong) Classes:

- Landslides
- Structural Debris
- Tree Uprooting

• Under-Represented (Weak) Classes:

  - Damaged Roads

  - Garbages

The Imbalance Problem: Training an ML network on this raw distribution will inject massive prediction bias. The network will develop high accuracy for landslides but completely fail to generalize or detect damaged road infrastructure, mistaking rare features for dominant backgrounds.

5. METHODOLOGICAL MITIGATION & LITERATURE REVIEW FOCUS

Per advisor directives, current work is strictly focused on conducting an exhaustive literature review targeting localized dataset balancing mechanisms. The research is moving along two parallel operational paths to solve the road and garbage data scarcity:

1. Algorithm-Level Adjustments (Loss Function Engineering):

Researching the implementation of "Focal Loss" and "Weighted Cross-Entropy Loss" models. These mathematical structures penalize the neural network exponentially more when it misclassifies minority groups (Damaged Roads), forcing the optimization algorithms to adjust network weights for sparse elements rather than ignoring them.

2. Data-Level Enhancements (Advanced Augmentation):

Reviewing papers on Class-Balanced Augmentation. While traditional transformations (rotations, color jitters) will be deployed, attention is being directed toward generative strategies to synthetically balance minority features while maintaining structural reality.

This dual-pronged strategy ensures that our unique, localized Sri Lankan fieldwork data forms a robust, un-biased foundation before launching baseline training runs in the upcoming weeks.